

# ROMAN KOSHKIN



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## SKILLS

**Programming languages:** Python (highest proficiency), C++, Matlab, R, HTML, JavaScript  
**ML Frameworks:** Pytorch (highest proficiency), HuggingFace, Lightning, scikit-learn, JAX  
**Frontend development:** React, Next.js  
**Virtualization, Infra, HPC:** Docker, Singularity, AWS, slurm  
**DB, LLMOps, MLOps, etc.:** Neo4j, Redis, MongoDB, LangSmith, LangChain, LangGraph, wandb, Apache Spark

## EDUCATION

**OIST** | *PhD (Machine Learning and Computational Neuroscience)* Okinawa, Japan  
Modeling memory in spiking networks and pattern detection  
**HSE University** | *Master (Cognitive Science)* Moscow, Russia  
with distinction, GPA: 8.9/10  
**VUMO University** | *Bachelor (Applied Linguistics)* Moscow, Russia  
with honors, GPA: 4.9/5

## WORK EXPERIENCE

**SB Intuitions, SoftBank** Tokyo, Japan  
*Research Scientist @ Speech Understanding Team* 07/2025 - current  
• Developing simultaneous speech-to-speech machine translation systems  
**OIST** Okinawa, Japan  
*Postdoctoral Fellow @ Neural Coding and Brain Computing Unit* 01/2025 - 06/2025  
• Developed simultaneous speech-to-speech machine translation systems.  
**Amazon** Tokyo, Japan  
*Research Scientist Intern @ JP Consumer Innovation* 09/2024 - 11/2024  
• Built multi-agent systems for open-ended automated market research.  
**NAIST** Nara, Japan  
*Special Research Intern @ NLP Group* 07/2023 - 10/2023  
• Developed speech-to-text SiMT models leveraging open-source causal LLMs  
**Araya** Tokyo, Japan  
*Research Intern @ Reinforcement Learning Team* 07/2023 - 10/2023  
• Developed robust EEG feature extractors to enable robot control via a brain-machine interface  
**HSE University** Moscow, Russia  
*Junior Research Fellow @ Center for Bioelectric Interfaces* 09/2017 - 07/2019  
• Coordinated a team of 3 researchers. Conceptualized research agenda, implemented EEG data collection and pre-processing pipelines, built ML models and other software, co-authored RF Patent 2747571

## SELECTED PUBLICATIONS

**Koshkin, R.**, Haesung, J., Liu, L., Shi, H., Zhao, M., Fujita, Y., Sudo, Y. (2026). Streaming Translation and Transcription Through Speech-to-Text Causal Alignment. *arXiv*. [LINK]  
**Koshkin, R.**, Dai, P., Fujikawa, N., Togami, M., Visentini-Scarzanella, M. (2025). MaRGen: Multi-Agent LLM Approach for Self-Directed Market Research and Analysis. *LLM For E-Commerce Workshop, KDD 2025*. [LINK]  
Yang, Z., Wei, L., **Koshkin, R.**, Chen, X., Nakamura S. (2025). SASST: Leveraging Syntax-Aware Chunking and LLMs for Simultaneous Speech Translation. *arXiv 2025*. [LINK]  
**Koshkin, R.**, Sudoh, K., Nakamura, S. (2024). LLMs Are Zero-Shot Context-Aware Simultaneous Translators. *EMNLP 2024*. [LINK]  
**Koshkin, R.**, Sudoh, K., Nakamura, S. (2024). TransLLaMa: LLM-based Simultaneous Translation System. *EMNLP 2024*. [LINK]  
**Koshkin, R.**, Fukai, T. (2024). convSeq: Fast and Scalable Method for Detecting Patterns in Spike Data. In *ICML 2024*. [LINK]

## AWARDS, GRANTS AND FELLOWSHIPS

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**OIST Teaching Fellowship**, 2023, ¥1.2M

**KAKENHI Grant-in-Aid**, 2023, ¥1.8M [LINK]

**Japan Society for the Promotion of Science Fellowship**, 2023 [LINK]

**Google PhD Fellowship**, 2021, \$10K [LINK]

## PATENTS

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**RF Patent 2747571**. EEG method for estimating listeners' reaction to audio content. [LINK]

## PROJECTS

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**toLLMatch** [LINK] 02/2024 - 05/2024

- Multilingual LLM-based speech-to-text simultaneous machine translation with no costly training

**TransLLaMa** [LINK] 07/2023 - 10/2023

- LLM-based speech-to-text simultaneous machine translation

**convSeq** [LINK] 01/2023 - 03/2023

- Fast and scalable convolution-based method for unsupervised detection of patterns in neural recordings

**SoNNet** [LINK] 09/2020 - 06/2024

- High-performance C++ library with a configurable user-friendly Python API for building spiking neural networks

**graphSeq** [LINK] 09/2022 - 12/2022

- Graph neural network-based method for embedding and clustering of neural spiking patterns

**M/EEG-based zero-shot speech decoding** [LINK] 09/2022 - 12/2022

- Re-implementation of an algorithm that decodes speech from human brain recordings (M/EEG) 0-shot

**Teaching Object Handling to a Robot** [LINK] 04/2020 - 09/2020

- Trained a robot to perform reach-and-grasp tasks by combining learned motor primitives

**Backpropagation-free learning for classification tasks** [LINK] 1/2020 - 4/2020

- Built a spike-timing dependent plasticity-based spiking neural network for image classification

**Neurobarometer** [LINK] 10/2017 - 07/2019

- Software & algorithm for EEG-based neuromarketing and consumer behavior research

## BOOK CHAPTERS AND OTHER PUBLICATIONS

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**Koshkin, R.**, Fukai, T. (2023). Unsupervised Detection of Cell Assemblies with Graph Neural Networks. In *ICLR 2023*. [LINK]

**Koshkin, R.**, Fukai, T. (2024). Astrocyte Regulation of Synaptic Plasticity Balances Robustness and Flexibility of Cell Assemblies. In *bioRxiv*. [LINK]

García, A. M., **Koshkin, R.**, de Oliveira Paiva, T. (2024). Electroencephalography. In A. M. Rojo López & R. Muñoz Martín (Eds.), *Research Methods in Cognitive Translation and Interpreting Studies* (pp. 279-303). John Benjamins Publishing Company. [LINK]

**Koshkin, R.**, Ossadtchi, A. (2017). Working Memory Load in Simultaneous Language Interpretation: An ERP Study. In *Proc. of the 4th Conference "Cognitive Science in Moscow: New Research"*. July 15, 2017, Moscow, Russia. p. 434 [LINK]

Garcia, A., **Koshkin, R.**, Paiva, T. (2023). EEG In Cognitive Translation and Interpreting Studies. (In review)

**Koshkin, R.**, Shtyrov, Y. , Myachykov, A, & Ossadtchi, A. (2018). Testing the Efforts Model of Simultaneous Interpreting. *PLoS ONE* 13(10): e0206129. [LINK]

**Koshkin, R.**, & Ossadtchi, A. (2017). Commentary: Functional Connectivity in the Left Dorsal Stream Facilitates Simultaneous Language Translation: An EEG Study. *Front. in Hum. Neurosci.*, 11(2), 273. [LINK]

**Koshkin, R.**, Ossadtchi, A. & Shtyrov, Y. (2017). Attention, Working Memory And Listening In Simultaneous Interpreting. *Russian J. of Cognitive Sci.*, 4(4). [LINK]

**Koshkin R.** (2016). Comparative Analysis of English-Russian and Russian-English Simultaneous Interpreting. *Bulletin of Moscow University, Series 22: Theory of Translation*. Vol. 2, 28-43 [LINK]

**Reviewer** for *Language, Cognition and Neuroscience, ICLR 2023, ICML 2026, ACL 2025, IEEE Transactions on Neural Networks and Learning Systems*.

**Science Mentor** Introduction to Deep Learning with Python, Okinawa, Japan

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POSTER PRESENTATIONS

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**Koshkin, R.**, Fukai, T. (2022). Astrocytes facilitate self-organization and remodeling of cell assemblies under STP-coupled STDP. *SfN Conference*, Nov 14-16, San Diego. [LINK]

**Koshkin, R.**, Shtyrov, Y. & Ossadtchi, A. (2017). Testing One Aspect of the Efforts Model of Simultaneous Interpreting: An ERP Study. In *Proceedings of the Workshop "Neurobiology Of Speech And Language"*, Oct. 27-29, 2017, SPb, Russia [LINK]

**Koshkin, R.**, Ossadtchi, A. & Shtyrov, Y.(2016). N1 ERP As an Index of Depth of Processing In Simultaneous Interpreting. In *Proceedings of Communication, Computation, and Cognitive Processes*, Sept. 28-29, 2016, Moscow, Russia [LINK]

**Koshkin, R.**, Ossadtchi, A. & Shtyrov, Y.(2017). Working Memory Load In Simultaneous Language Interpretation: An ERP Study. *IEEE International Symposium «Video and Audio Signal Processing»*, Jun. 26-30, 2017, SPb, Russia [LINK]